

Dheeraj Kadadalamari Bayareddy

Aspiring Machine Learning Engineer | MSc Data Science | [portfolio](#)

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PROFILE

MSc Data Science student transitioning into machine learning, bringing 3+ years of professional software engineering experience from a live insurance platform. Hands-on with Python, TensorFlow/Keras, Scikit-learn, and Large Language Models through self-directed projects in medical imaging and healthcare analytics — applying the production engineering discipline built over three years of API design, database architecture, and shipping reliable software under real deadlines. Seeking data-driven and machine learning roles in the European market.

SKILLS

- **Data & Machine Learning:** Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, deep learning (CNNs, transfer learning), feature engineering, classification, clustering, model evaluation, LLMs (learning)
- **Databases & Data Engineering:** SQL Server, MySQL schema design, stored procedures, query optimization, data pipeline design
- **Backend & APIs:** C#, ASP.NET Core, RESTful API design, microservices
- **Tools & Collaboration:** Jira, Git, Postman, structured logging, Agile/async remote collaboration

DATA / ML PROJECTS

Diabetic Retinopathy Severity Grading (Deep Learning)

2026

Python · TensorFlow/Keras · Gradio

- Built a 5-class DR severity grading system on the APTOS 2019 dataset (3,662 retinal images); trained and compared a custom CNN, ResNet50, and MobileNetV2 (transfer learning).
- Best results: ResNet50 QWK 0.67, custom CNN 67.9% accuracy, MobileNetV2 0.89 macro-AUC; used Ben Graham preprocessing, class weights, and Grad-CAM; deployed as a 3-model ensemble in a live Gradio demo.

Healthcare Fraud Detection (Machine Learning)

2026

Python · Scikit-learn · Pandas

- Compared 6 classification algorithms (Logistic Regression, Decision Tree, KNN, Random Forest, Gradient Boosting, SGD) on a synthetic health-insurance claims dataset; Random Forest gave the strongest fraud/legitimate separation.
- Feature analysis identified provider-patient distance and procedure count as the strongest fraud indicators — directly usable for a rules-based pre-screen.

Multiclass Pathology Classification in Voice Signals

2022

Python · Deep Learning

- Built an end-to-end audio-classification pipeline (preprocessing, feature extraction, model training, evaluation) across multiple pathology classes.

WORK EXPERIENCE

Software Engineer, C#/.NET Backend

Nov 2022 – Feb 2026

Paktolus (client: Bolt Insurance), Mumbai, India — Remote

- Built and maintained REST APIs and SQL Server data logic end-to-end for 3 insurance platform modules with no dedicated data-layer owner, including schema design and query optimization on large, time-sensitive datasets.
- Automated manual batch processes (policy renewals, document generation) via Windows Services, using structured logging for systematic root-cause debugging.

Full-Stack Web Development Intern

Sep 2021 – Oct 2021

Tequed Labs Pvt. Ltd., India

- Delivered backend and frontend tasks independently in a 1-month internship — server-side logic, API integration, and database interactions.

EDUCATION

MSc Data Science

Mar 2026 – Present

University of Europe for Applied Sciences, Potsdam, Germany

Coursework: Machine Learning · Marketing Analytics · Data Engineering · Cloud Computing · Data Visualization

Bachelor of Engineering (B.E.), Computer Science & Engineering

2018 – 2022

RajaRajeswari College of Engineering, Bangalore, India

Coursework: Python · SQL · Java · C · DSA · Databases · Software Engineering · Final-year project: voice-signal pathology classification (deep learning)

LANGUAGES

English (Professional Proficiency) · German (A2, learning)

INTERESTS

Applied Machine Learning · Data-Driven Strategy · LLMs & Generative AI · Continuous Learning